Nisha Jain

Embedded Firmware Engineer

RESUME_1484

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Total Experience: 4 years

PROFESSIONAL SUMMARY

Embedded Firmware Engineer with 4 years of experience in embedded firmware, using C, C++, ARM Cortex-M, FreeRTOS. Experienced in real-time systems, hardware debugging and measurable delivery outcomes.

TECHNICAL SKILLS

Skills: FreeRTOS, Oscilloscope, JTAG, UART, C++, Embedded Linux, STM32, ARM Cortex-M, CAN, C, SPI, I2C

Methods: Real-time systems, Hardware debugging, Protocol validation

PROFESSIONAL EXPERIENCE

Embedded Firmware Engineer | Asterline Engineering | Jan 2024 - Present

- Improved battery runtime by 22% with peripheral and sleep-state optimization
- Lowered field-reset incidents by 33% with watchdog recovery and diagnostics
- Reduced embedded task jitter by 26% using interrupt priority and queue tuning
- Coordinated engineering design reviews, acceptance criteria and test evidence with cross-functional stakeholders.

Associate Embedded Firmware Engineer | DeltaPoint Innovations | Jul 2022 - Dec 2023

- Reduced embedded task jitter by 38% using interrupt priority and queue tuning
- Validated 817 hardware test vectors across protocol and timing corner cases

PROJECTS

Bootloader Update Utility | SPI, Embedded Linux, CAN, C++

- Developed verified firmware transfer and rollback checks.
- Documented acceptance criteria and evaluated the design using hardware debugging and protocol validation.

Motor Control Interface | JTAG, C, C++, CAN

- Implemented CAN message handling, fault logging and deterministic state transitions.
- Documented acceptance criteria and evaluated the design using hardware debugging and protocol validation.

EDUCATION

B.Tech in Electronics and Communication Engineering, Bayview Institute of Engineering, 2022
Relevant Methods: Real-time systems, Hardware debugging, Protocol validation

CERTIFICATIONS

ARM Cortex-M Workshop (2026)